From Numbers to Perception: Multi-Band EDC Prediction

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Abstract

We predict **multi-band energy decay curves** (EDCs) from room geometry and frequency-dependent absorption, then reconstruct listening-ready room impulse responses (RIRs). A 1D-ConvNet decoder with a **log-domain slope penalty** replaces our previous LSTM: **90 M → 9 M parameters**, 24 one-third-octave bands (100 Hz–20 kHz), and T_{30} within the 5% JND for most rooms. github.com/TUIlmenauAMS/LSTM-Model-Energy-Decay-Curves

9 M parameters — 90% smaller than LSTM

24 bands 1/3-octave — 100 Hz – 20 kHz

T_{30} $R^2 = 0.90$ — MAE 0.05 s, mostly within 5% JND

5× faster inference — real-time VR / auralization

Why predict EDCs?

- Full RIRs are high-dimensional; waveform synthesis is costly and often non-physical.
- The Schroeder EDC is compact and already determines T_{30}, EDT, C_{50}.
- **Pipeline:** geometry + $\alpha(f)$ → multi-band EDC → RSS polarity → auralizable RIR.

Room features → **ConvNet** → **EDC**(f,t) → **RIR**

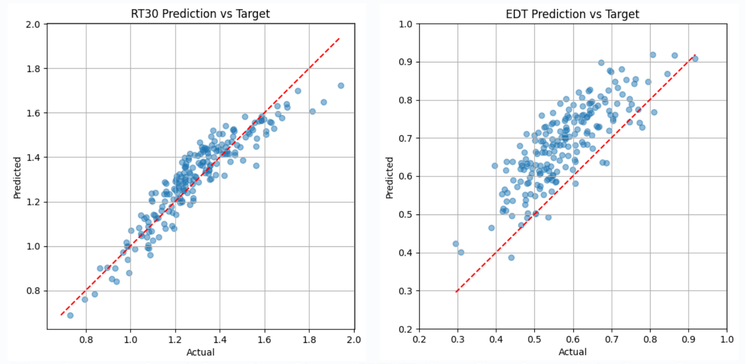
Log-domain loss with slope penalty

Train in dB so the loss matches perceived decay:

$$y_{\mathrm{dB}} = 10\log_{10}(y+\varepsilon)$$
$$\mathcal{L}_t = \mathrm{MSE}(\hat{y}_{\mathrm{dB}}, y_{\mathrm{dB}}) + \alpha\,\mathrm{MSE}(\Delta\hat{y}_{\mathrm{dB}}, \Delta y_{\mathrm{dB}})$$
$$\Delta y_{\mathrm{dB}}[n] = y_{\mathrm{dB}}[n+k] - y_{\mathrm{dB}}[n]$$

$k = 50$ samples, $\alpha = 0.2$. The slope term learns the local decay rate (hence T_{30}/EDT) and suppresses non-monotonic wiggles.

T_{30} and EDT on the test set

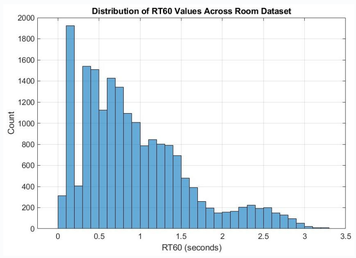


T_{30} hugs the identity line; EDT (early slope) is harder.

Dataset — 6 000 shoebox rooms

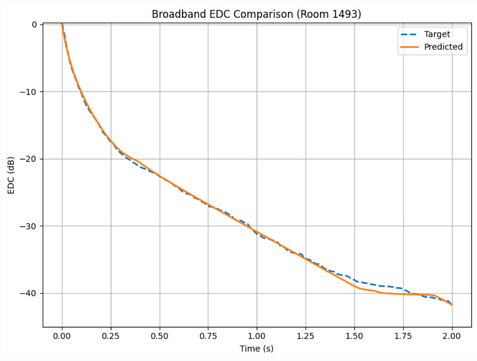
Pyroomacoustics, 48 kHz. Inputs: L, W, H, source/receiver (x, y, z), wall $\alpha(f)$.

Parameter	Range
Dimensions $L \times W \times H$	3–6 × 3–6 × 2.5–4 m
Source–receiver distance	1–4 m
Absorption $\alpha(f)$	0.14–0.65, 100 Hz–20 kHz
Rooms / bands	6 000 / 24



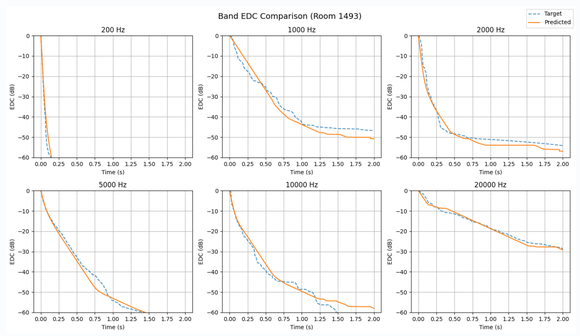
T_{60} peaks at everyday rooms, with a reverberant tail.

Predicted vs. target EDC



Held-out room: prediction tracks the Schroeder target through the late tail.

Multi-band EDCs (24 × 1/3-octave)



Six of 24 1/3-octave bands, same room. Some frequencies die faster, as materials predict — a broadband EDC cannot show this.

ConvNet vs. prior LSTM

SANE 2025 LSTM: ∼90 M params, broadband. **Now:** MLP encoder + 1D-CNN decoder, **9 M** params, 24 bands.

Input — geometry, src/rcv, $\alpha(f)$

▼

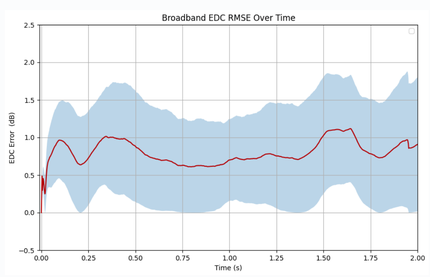
MLP encoder — latent + band

▼

1D-CNN + interpolate → EDC

Interpolation removes the LSTM staircase.

EDC error and parameters



Mean RMSE ≈ 0.7–1.1 dB over 2 s (shaded: ±1 std.).

Parameter	RMSE	MAE	R^2
EDT (s)	0.10	0.07	0.79
T_{20} (s)	0.06	0.04	0.93
T_{30} (s)	0.07	0.05	0.90
C_{50} (dB)	0.47	0.35	0.67

T_{30} mostly within 5% JND. EDT MAE rose vs. broadband LSTM (0.033→0.07 s) in exchange for 24 bands.

From EDC to RIR

Differentiate the EDC → magnitude; **RSS** polarity ($p = 0.9$) keeps LF coherence.

$$d[n] = -(\mathrm{EDC}[n+1] - \mathrm{EDC}[n])$$
$$h_{\mathrm{mag}}[n] = \sqrt{\max(d[n], 0)}$$

Prior LSTM MUSHRA (N=12): RSS **85.8** vs. reference **95.3** (no significant difference). ConvNet EDCs are smoother, so we expect equal or better listening scores.

Takeaways, outlook & contact

Takeaways: multi-band EDCs from geometry; log-MSE + slope penalty; T_{30} within 5% JND at 9 M params. **Next:** non-shoebox rooms, measured RIRs, psychoacoustic loss. **Contact:** m.imran@tu-braunschweig.de · gerald.schuller@tu-ilmenau.de